

Supporting Information Appendix for “Whether richer countries are happier depends on the survey”

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S1 Data

S1.1 World Values Survey and European Values Study

The World Values Survey (WVS) is the longest-running cross-national survey of values and subjective well-being. I use the WVS time-series file (1), which pools the seven waves conducted between 1981 and 2022, and complete it with the Joint EVS/WVS 2017–2022 dataset (2), which adds the 36 national surveys of the 2017 wave of the European Values Study (EVS, conducted in 2017–2021) and 2 recent WVS surveys (India 2023, Uzbekistan 2022). The EVS asks the same well-being questions, and the combination of the two surveys is standard practice (the so-called Integrated Values Surveys). The EVS and the WVS are distinct surveys: when both surveyed the same country the same year (3 cases), I keep both surveys as separate observations. For brevity, I refer to the combined data as the WVS: 501,911 respondents in 342 country-years and 112 countries. National samples are designed to be representative of the adult population, typically with 1,000 to 3,000 respondents, and most interviews are conducted face-to-face (though some recent national surveys used web, mail or mixed modes, e.g. the WVS in the United States in 2017 and in Japan in 2019, or the EVS in Switzerland in 2017). All statistics use the survey weights provided by the WVS and the EVS.

The WVS contains two questions on subjective well-being:

- *Happiness*: “Taking all things together, would you say you are: *Very happy*; *Quite happy*; *Not very happy*; *Not at all happy*.”
- *Life satisfaction*: “All things considered, how satisfied are you with your life as a whole these days?”, answered on a scale from 1 (*Completely dissatisfied*) to 10 (*Completely satisfied*).

Non-responses are rare (1.4% for satisfaction and 1.7% for happiness on average) and uncorrelated with GDP per capita (correlation: -0.09); they are excluded.

Indicators of national well-being. Because there is no consensus on how to aggregate ordinal answers into a national indicator, and because the ranking of countries can depend on this choice (3), I compute eight indicators for each country-year:

- *Happy*: share answering *Quite* or *Very happy*;
- *Very Happy*: share answering *Very happy*;
- *Very Unhappy*: share answering *Not at all happy*;
- *V. Happy* – *V. Unhappy*: difference between the two previous shares;
- *Happiness (mean)*: mean happiness, with answers coded +3, +1, -1 , -3 ;
- *Satisfaction (mean)*: mean life satisfaction;
- *Satisfied*: share answering 6 to 10 to the life-satisfaction question;
- *Happy* + *Satisfied*: average of *Happy* and *Satisfied*, the index used by Inglehart and Klingemann (4).

The indicators are positively but imperfectly correlated: across country-years, the correlation between *Happy* and *Satisfied* is 0.72, and between *Very Happy* and *Satisfaction (mean)* it is only 0.60 (Table S7). For the life-satisfaction question, I also compute the shares answering 8–10, 9–10, 10, 1–4 and 1–2, for comparisons with Gallup.

S1.2 Gallup World Poll

The Gallup World Poll interviews around 1,000 adults per country and year in more than 140 countries, by telephone where coverage is high (typically in high-income countries) and face-to-face elsewhere (5). Its life-evaluation question is the Cantril ladder (6): “Please imagine a ladder, with steps numbered from 0 at the bottom to 10 at the top. The top of the ladder represents the best possible life for you and the bottom of the ladder represents the worst possible life for you. On which step of the ladder would you say you personally feel you stand at this time?”

I use the distribution of answers by country and wave, tabulated from the Gallup World Poll microdata (2,351 country-years, 168 countries, 2006–2023), from which I compute the mean and the shares answering 6–10, 8–10, 9–10, 10, 0–4 and 0–2. These distributions are unweighted. To cover the most recent years, I also use the ladder means published in the World Happiness Report 2026 (7), which average Gallup data over three years up to 2023–2025.

S1.3 Income

My benchmark income measure is the log of GDP per capita at purchasing power parity (PPP, constant 2021 international dollars) from the World Development Indicators (8). As an alternative, I use GDP per capita at market exchange rates (constant 2015 US dollars). Missing values are filled in automatically, in three steps: (i) PPP series, which start in 1990, are extended backwards (and gaps filled) using the growth rate of real GDP per capita; (ii) for countries absent from the World Bank data (e.g. Taiwan, or Venezuela in recent years), I use the IMF World Economic Outlook (9), converted into the constant dollars of the World Bank series with a U.S. price deflator (the ratio of World Bank to IMF values for the United States); (iii) remaining gaps are filled with the closest year available (Montenegro 1996). Territories without national accounts use the GDP of the closest entity (e.g. the United Kingdom for Northern Ireland). A robustness check excludes all imputed values.

S1.4 Regions

My main classification groups countries into the five UN regional groups (Figure S1): *Africa*, *Asia* (the Asia-Pacific group, which includes the Middle East and Central Asia), *Eastern Europe* (the former Eastern Bloc, excluding Central Asia), *Latin America* (including the Caribbean), and *Western countries* (Western Europe, North America, Australia, New Zealand, and Turkey). Territories that are not UN members (e.g. Taiwan, Hong Kong, Puerto Rico, Kosovo) are assigned to the group of their neighbors. It departs from the UN groups for two countries only: Israel, a member of the Western group since 2000 because it was denied access to the Asia-Pacific group, is classified in Asia, and Cyprus, a member of the Asia-Pacific group, is classified with Western countries.

For robustness, I use four alternative classifications: the exact UN regional groups (with Israel in the Western group and Cyprus in Asia); a six-region variant separating the Middle East and grouping Central Asia with Eastern Europe; the seven World Bank regions (East Asia and Pacific, Europe and Central Asia, Latin America and the Caribbean, Middle East and North Africa, North America, South Asia, Sub-Saharan Africa); and six continents (Africa, North America, Latin America and the Caribbean, Asia, Europe, Oceania). The last two isolate Latin America but group Eastern and Western Europe.

See Fig. S1.

S2 Income vs. region: method and additional results

For each well-being indicator and each income indicator, I estimate three linear regressions across country-years i :

$$well-being_i = \alpha_1 + \beta_1 income_i + u_i \quad (S1)$$

$$well-being_i = \alpha_2 + \gamma_2 region_i + e_i \quad (S2)$$

$$well-being_i = \alpha_3 + \beta_3 income_i + \gamma_3 region_i + \varepsilon_i \quad (S3)$$

where $region_i$ denotes the vector of region dummies. Denoting R_1^2 , R_2^2 and R_3^2 their coefficients of determination, the share of the variance explained by income alone is R_1^2 . To compare income and region, I decompose R_3^2 between the two regressors following Lindeman et al. (10) (LMG), i.e. using the Shapley value (11). With two (groups of) regressors, the LMG contribution of income is the average of its contribution when entered first (R_1^2) and when entered second ($R_3^2 - R_2^2$). The *share of explained variance due to income* is therefore

$$s = \frac{R_1^2 + (R_3^2 - R_2^2)}{2 R_3^2}, \quad (S4)$$

and region is the better predictor whenever $s < 50\%$.

One feature of this comparison could favor region: region has four degrees of freedom while log GDP has one. I address this concern in three ways. First, I use categorical income variables (sextiles and clusters) with as many or more categories as regions. Second, I compute s with adjusted R^2 , which penalizes additional regressors; with more than 300 observations, the adjustment is small and only reinforces the conclusions. Third, and more directly relevant to *predictive* capacity, I compute leave-one-country-out cross-validated R^2 : each country's observations are predicted by a model estimated on all other countries. Unlike the in-sample R^2 , the cross-validated R^2 decreases when a model has superfluous parameters, and it is negative when a model predicts worse than the sample mean. Other robustness checks include testing each WVS wave separately, excluding the EVS surveys, weighting countries by their population, keeping only the last observation of each country, excluding the pandemic years 2020–2021, excluding imputed GDP values, using the previous vintage of GDP data (PPP in constant 2017 dollars), alternative region classifications, and excluding Latin America or Eastern Europe.

Table S5 shows that the region advantage rests on two well-known “anomalies”: Latin America, happier than its income would predict (12, 13), and Eastern Europe, unhappier (14). Region remains the better predictor when either one is excluded (in 70% of cases without Latin America and 73% without Eastern Europe), but not when both are (27%). Consistently, the result holds with classifications that isolate both groups (exact UN groups and six regions: 100%), and weakens or disappears with the World Bank regions (55%) and the six continents (48%), which isolate Latin America but merge Eastern and Western Europe.

The cross-sectional relation should not be confused with the relation over time. Within countries (country fixed effects, 312 observations), mean satisfaction increases with log GDP per capita, with a slope of 2.31 per unit of $\log_{10}$ GDP, larger than the cross-sectional slope (1.14), although income growth explains only 0.34 of the within-country variance (Table S9). These associations are consistent with Stevenson and Wolfers (15) and Sacks et al. (16) rather than with a strict version of the Easterlin paradox (17, 18), although they may reflect common shocks (e.g. transition recessions) rather than the effect of income per se. Across countries, Deaton (19) found that, controlling for the level of GDP per capita, growth was negatively associated with life satisfaction in the 2006 Gallup World Poll. I do not find this: controlling for log GDP per capita, one percentage point of annual growth over the previous five years is associated with 0.039 points of mean satisfaction in the WVS (s.e. 0.019, $p = .044$) and 0.001 points of mean ladder in Gallup 2006–2023 (s.e. 0.010, $p = .957$), with standard errors clustered by country.

Additional results are reported in the following tables and figures: the share of explained variance due to income for each indicator and income measure (Table S2; Gallup: Table S6), the slopes of each

indicator on log GDP (Table S8), other country-level correlates (Table S10), the countries most often ranked happiest (Table S11), and scatter plots of each indicator against GDP per capita (Figs. S2 and S4–S10).

S3 Survey experiment

Gallup and the WVS differ in many respects. Gallup asks the Cantril ladder on a 0–10 scale, the WVS life satisfaction on a 1–10 scale. Gallup relies on a centralized, standardized design, with about 1,000 respondents per country every year, interviewed by telephone in high-income countries and face-to-face elsewhere; the WVS is fielded every five years or so by national teams, mostly face-to-face, with sampling designs, modes and survey years that vary across countries. These differences fall into two categories that could explain why the two datasets disagree: the question (wording and scale) and everything else—sampling frames, interview modes, questionnaire context, translations, and survey years, which I refer to as *samples* in a broad sense. To separate the two, I embedded a randomized experiment in an original survey (20). The survey was conducted online between April 15 and July 3, 2025, by Bilendi (or Kantar in Saudi Arabia) on 11,000 respondents in 10 high-income countries: the United States (3,000), Japan (2,000), Saudi Arabia (1,000), Germany (1,048), the United Kingdom (826), France (798), Italy (756), Spain (603), Poland (500) and Switzerland (469); the survey also covered Russia, where the well-being question was censored. Samples were stratified with quotas on gender, age, income, education, region and urbanicity (as well as race in the U.S. and citizenship in Saudi Arabia), and results are reweighted to match population frequencies along these dimensions (with weights trimmed between 0.25 and 4).

Toward the end of the questionnaire, after questions on global policies, each respondent was randomly assigned to one of four versions of the life-evaluation question:

- *Ladder 0–10* (Gallup’s original): the Cantril ladder question quoted in Section S1.2, with steps numbered from 0 (*Worst possible*) to 10 (*Best possible*);
- *Ladder 1–10*: same wording, with steps numbered from 1 to 10;
- *Satisfaction 0–10*: the WVS question quoted in Section S1.1, with answers from 0 (*Completely dissatisfied*) to 10 (*Completely satisfied*);
- *Satisfaction 1–10* (WVS’s original): same wording, with answers from 1 to 10.

This 2×2 design identifies the effect of wording (ladder vs. satisfaction) and scale (0–10 vs. 1–10), and their interaction, holding the sample, mode, context and time constant. Within each country, each branch receives around 25% of respondents.

Although levels matter less than correlations for our question, the experiment also explains why the WVS yields higher levels than Gallup. Table S12 reports regressions of the answer on the randomized treatments, with country fixed effects, survey weights and heteroskedasticity-robust standard errors. The ladder wording shifts answers by -0.45 points on average (s.e. 0.05): respondents rate their lives lower on the “best possible life” ladder than when asked how satisfied they are, in line with evidence that the ladder evokes power and wealth more than life satisfaction does (21). Answers on the 1–10 scale are 0.14 points higher than on the 0–10 scale (s.e. 0.05). Once 1–10 answers are linearly stretched to 0–10 with $x \mapsto (x - 1) \times 10/9$, which maps 1 to 0 and 10 to 10, the scale effect becomes -0.26 : the stretch lowers each answer x by $(10 - x)/9$, i.e. by about 0.4 points at the average answer. Wording and scale do not interact. The wording effect varies markedly across countries (Figure S3; $\chi^2(9) = 93.4$, $p < .001$). Japan is the exception: it is the only country where the ladder yields *higher* answers than the satisfaction question (by 0.36 points); elsewhere, the wording effect ranges from -1.12 (Spain) to -0.34 points. Finally, the ladder does not steepen the income gradient *within* countries: the interaction between the ladder wording and the respondent’s income quartile is -0.004 ($p = .919$), against a gradient of 0.69 points per quartile.

To relate these effects to the Gallup–WVS gap, I define for each country c the observed gap $D_c = W_c - G_c$, where W_c is mean satisfaction in the WVS and G_c the mean ladder in Gallup the same year (as in the main text), and the question effect $Q_c = F_c(\text{satisfaction } 1\text{--}10) - F_c(\text{ladder } 0\text{--}10)$, the difference between the mean answers to the two original variants in the experiment, measured on the same sample. The residual $R_c = D_c - Q_c$ is the part of the gap due to samples in the broad sense. On average, WVS answers exceed Gallup answers by $D = 0.60$ points, and the question effect is even larger: $Q = 0.71$ points (Tables S13 and S14). The question thus fully accounts for the higher level of WVS answers in high-income countries, and the residual is small ($R = -0.11$, 95% CI $[-0.26; 0.04]$). Since the two components move in opposite directions, another way to summarize the level difference is to compare the size of the two movements: the question accounts for $|\bar{Q}| / (|\bar{Q}| + |\bar{R}|) = 0.86$ (95% CI $[0.76; 0.99]$) of the gross difference in levels.

The question effect does not, however, explain why the gap varies across countries. Since $D_c = Q_c + R_c$, the cross-country variance of the gap splits into $\text{var}(D) = \text{cov}(D, Q) + \text{cov}(D, R)$. The share attributable to the question, $\text{cov}(D, Q) / \text{var}(D)$, is only 0.11 (95% CI $[-0.33; 0.55]$): countries with a larger question effect do not have a larger gap. Accordingly, predicting each country’s Gallup mean by subtracting its own question effect Q_c from its WVS mean does worse (root mean squared error: 0.54) than simply subtracting the average gap (0.35). Samples asked the same question also differ markedly. Gallup respondents (2023–2025) rate their lives 0.75 points higher on the ladder 0–10 than online respondents in 2025, and WVS respondents 0.75 points higher on satisfaction 1–10 (even in the United States, where the 2017 WVS was itself conducted online: 0.49 points). And in the 10 countries surveyed separately by the EVS and the WVS in 2017–2022, with the same question, mean satisfaction differs by 0.28 points on average (up to 0.53).

S4 Representativeness of the surveys

If samples matter, which dataset is closer to the truth? There is no clear-cut argument in favor of either dataset, nor of the online sample used in the experiment. Gallup applies a centralized, standardized design: probability samples of about 1,000 adults per year, drawn by random-digit dialing where telephone coverage is high and by multistage area sampling elsewhere, with random selection of the respondent within the household and post-stratification weights on gender, age and, where possible, education (22). But its samples are small; its face-to-face samples select households by random route, with replacement of non-responding households; response rates are not published; interviews switched from face-to-face to telephone during the pandemic; and in 2023, a fifth of the interviews in 27 high-income countries came from web panels, most of them opt-in (22). Its subgroup estimates have also been questioned (23). Besides, Gallup microdata are not public, so that the representativeness of its samples cannot be assessed independently; lacking quotas, its samples could under-represent hard-to-reach groups such as less educated or rural people. The WVS requires probability samples of at least 1,200 respondents but allows quota selection at the last stage, and its sampling designs, modes and survey years vary across countries (24).

I test the representativeness of WVS/EVS samples directly (Table S15). Compared with population benchmarks (8, 25), WVS samples over-represent tertiary-educated adults (aged 25 and over) by a median factor of 1.98 in countries with a GDP per capita below 10,000 dollars, against 1.05 in countries above 30,000 dollars (295 surveys). They also over-represent urban residents in poorer countries (median ratio 2.19 vs. 1.22) and under-represent people aged 65 or more (0.64 vs. 1.01). The internal criterion of Kohler (26)—women should make up half of married or cohabiting respondents—is violated at the 5% level in 36% of surveys. WVS samples are thus less representative in poorer countries, as noted by Deaton (19), which is where their gap with Gallup is largest. However, correcting these observable imbalances has little effect. Controlling for log GDP, the over-representation of the tertiary-educated does not predict the gap (coefficient -0.07 , s.e. 0.07 , $p = .311$), and reweighting each WVS sample to the benchmark share

of tertiary-educated adults barely changes its relation with income: in the country-years also covered by Gallup, the R^2 of mean satisfaction on log GDP goes from 0.19 to 0.20, against 0.54 for Gallup. The discrepancy thus stems from unobserved differences in who responds, how, and in which context, which the available evidence does not allow me to attribute to one dataset rather than the other.

S5 Supplementary tables and figures

Table S1: Share of the variance of national well-being explained by income alone (R^2), WVS/EVS 1981–2023.

Well-being indicator	log GDP p.c.		Sextile PPP	Income cluster				Region
	PPP	nominal		$k=5$ PPP	$k=6$ PPP	$k=7$ PPP	$k=7$ nominal	
Happy	0.14	0.20	0.20	0.19	0.20	0.20	0.23	0.30
Very Happy	0.00	0.00	0.03	0.03	0.04	0.05	0.04	0.33
Very Unhappy	0.07	0.10	0.12	0.10	0.13	0.13	0.14	0.14
V. Happy – V. Unhappy	0.00	0.01	0.04	0.04	0.04	0.05	0.05	0.33
Happiness (mean)	0.04	0.08	0.10	0.08	0.09	0.10	0.11	0.33
Satisfaction (mean)	0.23	0.29	0.23	0.21	0.22	0.23	0.29	0.41
Satisfied	0.32	0.38	0.32	0.30	0.32	0.32	0.38	0.39
Happy + Satisfied	0.27	0.35	0.30	0.28	0.30	0.30	0.35	0.38
Mean	0.14	0.18	0.17	0.15	0.17	0.17	0.20	0.33
Max	0.32	0.38	0.32	0.30	0.32	0.32	0.38	0.41

Note: R^2 of the regression of each well-being indicator (rows) on each income indicator (columns), over 342 WVS/EVS country-years (a few with missing GDP are excluded). The last column gives the R^2 of the regression on region dummies. The highest R^2 of each row is in bold. The replication files include an Excel version of this table with the R^2 of all region classifications (tables/main/r2_income_all_regions.xlsx).

Table S2: Share of explained variance due to income rather than region (LMG), WVS/EVS 1981–2023.

Well-being indicator	log GDP p.c.		Sextile PPP	Income cluster			
	PPP	nominal		$k=5$ PPP	$k=6$ PPP	$k=7$ PPP	$k=7$ nominal
Happy	0.29	0.36	0.38	0.35	0.37	0.38	0.41
Very Happy	0.00	0.01	0.07	0.06	0.08	0.10	0.10
Very Unhappy	0.28	0.38	0.43	0.40	0.47	0.47	0.50
V. Happy – V. Unhappy	0.01	0.02	0.09	0.07	0.09	0.11	0.11
Happiness (mean)	0.09	0.13	0.18	0.14	0.17	0.19	0.19
Satisfaction (mean)	0.32	0.38	0.32	0.29	0.31	0.32	0.38
Satisfied	0.43	0.49	0.43	0.41	0.43	0.44	0.49
Happy + Satisfied	0.39	0.46	0.42	0.39	0.41	0.42	0.47
Mean	0.23	0.28	0.29	0.26	0.29	0.30	0.33
Max	0.43	0.49	0.43	0.41	0.47	0.47	0.50

Note: Share of the R^2 of Equation (S3) attributed to income by the LMG (Shapley) decomposition, Equation (S4). Values below 0.5 indicate that region is the better predictor. Values above 0.5 (income is the better predictor) are in bold.

[\(Back to Section S2\)](#)

Table S3: Out-of-sample predictive capacity: leave-one-country-out cross-validated R^2 .

Data	Indicator	Obs.	log GDP PPP	Best income variable	Region	log GDP PPP + Region
WVS, all waves	Happy	340	0.12	0.18	0.26	0.32
WVS, all waves	Very Happy	340	−0.03	−0.04	0.27	0.27
WVS, all waves	Very Unhappy	340	0.06	0.10	0.10	0.11
WVS, all waves	V. Happy – V. Unhappy	340	−0.02	−0.02	0.27	0.26
WVS, all waves	Happiness (mean)	340	0.02	0.05	0.29	0.30
WVS, all waves	Satisfaction (mean)	340	0.21	0.23	0.37	0.46
WVS, all waves	Satisfied	340	0.30	0.34	0.36	0.50
WVS, all waves	Happy + Satisfied	338	0.26	0.31	0.35	0.47
WVS, last obs. per country	Happy	115	0.05	0.03	0.11	0.12
WVS, last obs. per country	Very Happy	115	0.01	−0.05	0.17	0.16
WVS, last obs. per country	Very Unhappy	115	0.04	0.06	0.01	0.02
WVS, last obs. per country	V. Happy – V. Unhappy	115	−0.01	−0.08	0.16	0.15
WVS, last obs. per country	Happiness (mean)	115	−0.04	−0.09	0.12	0.10
WVS, last obs. per country	Satisfaction (mean)	115	0.18	0.19	0.30	0.33
WVS, last obs. per country	Satisfied	115	0.29	0.31	0.31	0.38
WVS, last obs. per country	Happy + Satisfied	115	0.24	0.25	0.28	0.34
Gallup, all years (2006–23)	Satisfaction (mean)	2344	0.61	0.66	0.49	0.68
Gallup, all years (2006–23)	Satisfied	2344	0.65	0.72	0.52	0.72
Gallup, all years (2006–23)	Unsatisfied (0/1–4)	2344	0.63	0.65	0.48	0.68
Gallup, last obs. per country	Satisfaction (mean)	167	0.60	0.58	0.45	0.64
Gallup, last obs. per country	Satisfied	167	0.65	0.68	0.52	0.70
Gallup, last obs. per country	Unsatisfied (0/1–4)	167	0.65	0.60	0.48	0.67
WHR, 2023–25 average	Satisfaction (mean)	147	0.62	0.60	0.52	0.69

Note: Each country's observations are predicted with a model estimated on all other countries. Cross-validated $R^2 = 1 - \sum(y - \hat{y}_{\text{out-of-sample}})^2 / \sum(y - \bar{y})^2$; it can be negative when a model predicts worse than the overall mean. In each row, the highest cross-validated R^2 among log GDP PPP, the best income variable and region is in bold.

Table S4: Robustness: income vs. region across specifications and datasets.

Specification	Obs.	Countries	R^2 income		R^2 region	Share of explained variance due to income		Region better (share of cases)
			log PPP	best		log PPP	best	
WVS, all waves	340	112	0.14	0.20	0.33	0.23	0.33	0.98
WVS, waves 1–2 (1981–91)	26	21	0.07	0.43	0.58	0.09	0.40	1.00
WVS, wave 3 (1995–99)	55	55	0.16	0.33	0.65	0.14	0.28	1.00
WVS, wave 4 (1999–2004)	40	40	0.19	0.31	0.33	0.27	0.47	0.43
WVS, wave 5 (2004–09)	58	58	0.21	0.35	0.49	0.22	0.38	0.98
WVS, wave 6 (2010–16)	60	60	0.08	0.22	0.27	0.14	0.41	0.96
WVS, wave 7 (2017–23)	102	92	0.09	0.17	0.28	0.21	0.35	1.00
WVS, wave 7 w/o 2020–21	80	72	0.08	0.19	0.29	0.19	0.38	0.95
WVS, population-weighted	340	112	0.10	0.19	0.21	0.25	0.44	0.77
WVS, last obs. per country	115	112	0.13	0.19	0.25	0.26	0.39	0.88
WVS, no GDP imputation	321	108	0.13	0.20	0.32	0.22	0.33	0.98
WVS, GDP PPP 2017 \$ (old)	340	112	0.14	0.20	0.33	0.24	0.33	0.98
WVS, without EVS surveys	304	108	0.12	0.19	0.37	0.18	0.28	1.00
Gallup, all years (2006–23)	2344	167	0.42	0.48	0.42	0.46	0.51	0.31
Gallup, last obs. per country	167	167	0.42	0.44	0.41	0.44	0.48	0.37
Gallup, last obs., WVS countries	109	109	0.44	0.47	0.43	0.46	0.49	0.35
Gallup, last obs., 6 regions	167	167	0.42	0.44	0.42	0.43	0.46	0.41
Gallup, last obs., WB regions	167	167	0.42	0.44	0.38	0.47	0.50	0.29
WHR, 2023–25 average	147	147	0.63	0.63	0.55	0.56	0.56	0.00

Note: Averages over well-being indicators (8 for the WVS; 7 satisfaction indicators for Gallup; mean ladder for the WHR). “Best” refers to the income measure with the highest average R^2 in the main WVS specification (Cluster k=7 nominal). “Region better” is the share of indicator $\times$ income-measure combinations for which $s < 0.5$; values above 0.5 are in bold. Alternative region classifications and exclusions of regions: see Table S5.

Table S5: Income vs. region by region classification and without Latin America or Eastern Europe (WVS/EVS).

Region classification	Groups	Obs.	R^2 income (log PPP)	R^2 region	Share due to income (log PPP)	Region better (share of cases)
5 UN regional groups (main)	5	340	0.14	0.33	0.23	0.98
Exact UN regional groups	5	340	0.14	0.33	0.23	1.00
6 regions (Middle East separate)	6	340	0.14	0.32	0.22	1.00
World Bank regions	7	340	0.14	0.19	0.37	0.55
Continents	6	340	0.14	0.17	0.40	0.48
Main, w/o Latin America	4	285	0.15	0.30	0.26	0.70
Main, w/o Eastern Europe	4	256	0.16	0.22	0.29	0.73
Main, w/o both	3	201	0.18	0.20	0.35	0.27

Note: Averages over the eight well-being indicators. R^2 and share of explained variance due to income (Equation (S4)) with log GDP p.c. (PPP). “Region better”: share of the indicator $\times$ income-measure combinations for which $s < 0.5$; values above 0.5 are in bold. Region classifications are described in Section S1.4.

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Table S6: Share of explained variance due to income rather than region (LMG), Gallup, last observation per country.

Well-being indicator	log GDP p.c.		Sextile PPP	Income cluster			
	PPP	nominal		$k=5$ PPP	$k=6$ PPP	$k=7$ PPP	$k=7$ nominal
Satisfied	0.58	0.61	0.59	0.58	0.60	0.59	0.60
Very satisfied (8–10)	0.44	0.51	0.49	0.47	0.49	0.51	0.53
Extremely satisfied (9–10)	0.16	0.25	0.24	0.27	0.22	0.30	0.28
Completely satisfied (10)	0.02	0.02	0.08	0.09	0.06	0.08	0.08
Unsatisfied (0/1–4)	0.61	0.60	0.60	0.59	0.61	0.60	0.59
Dissatisfied (0/1–2)	0.66	0.63	0.67	0.66	0.67	0.67	0.65
Satisfaction (mean)	0.60	0.62	0.59	0.58	0.60	0.59	0.60
Mean	0.44	0.46	0.47	0.46	0.47	0.48	0.48
Max	0.66	0.63	0.67	0.66	0.67	0.67	0.65

Note: 168 countries, last Gallup observation available (2006–2023). Same definitions as Table S2. Values above 0.5 (income is the better predictor) are in bold.

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Table S7: Correlation between national well-being indicators (WVS country-years).

	Happy	Very Happy	Very Unhappy	V. Happy – V. Unhappy	Happiness (mean)	Satisfaction (mean)	Satisfied	Happy + Satisfied
Happy	1.00	0.60	–0.73	0.70	0.89	0.73	0.72	0.90
Very Happy	0.60	1.00	–0.37	0.98	0.89	0.60	0.51	0.59
Very Unhappy	–0.73	–0.37	1.00	–0.54	–0.67	–0.58	–0.57	–0.69
V. Happy – V. Unhappy	0.70	0.98	–0.54	1.00	0.94	0.66	0.58	0.67
Happiness (mean)	0.89	0.89	–0.67	0.94	1.00	0.75	0.69	0.83
Satisfaction (mean)	0.73	0.60	–0.58	0.66	0.75	1.00	0.96	0.93
Satisfied	0.72	0.51	–0.57	0.58	0.69	0.96	1.00	0.95
Happy + Satisfied	0.90	0.59	–0.69	0.67	0.83	0.93	0.95	1.00

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Table S8: Slope of national well-being indicators with respect to $\log_{10}$ GDP per capita (PPP).

Data	Indicator	Slope	s.e.	<i>p</i> -value	R^2	Obs.
WVS	Happy	0.109	0.019	0.000	0.144	340
WVS	Very Happy	0.002	0.025	0.948	0.000	340
WVS	Very Unhappy	−0.022	0.004	0.000	0.072	340
WVS	V. Happy – V. Unhappy	0.023	0.026	0.379	0.004	340
WVS	Happiness (mean)	0.264	0.083	0.002	0.044	340
WVS	Satisfaction (mean)	1.143	0.142	0.000	0.233	340
WVS	Satisfied	0.236	0.024	0.000	0.316	340
WVS	Happy + Satisfied	0.173	0.020	0.000	0.275	338
Gallup	Satisfied	0.372	0.017	0.000	0.655	2344
Gallup	Very satisfied (8–10)	0.218	0.018	0.000	0.442	2344
Gallup	Extremely satisfied (9–10)	0.065	0.009	0.000	0.178	2344
Gallup	Completely satisfied (10)	0.012	0.005	0.008	0.015	2344
Gallup	Unsatisfied (0/1–4)	−0.304	0.012	0.000	0.639	2344
Gallup	Dissatisfied (0/1–2)	−0.135	0.009	0.000	0.432	2344
Gallup	Satisfaction (mean)	1.779	0.083	0.000	0.611	2344

Note: OLS across country-years; standard errors clustered by country. Gallup: 2006–2023.

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Table S9: Within-country relation between national well-being and $\log_{10}$ GDP per capita (country fixed effects, WVS).

Indicator	Slope	s.e.	<i>p</i> -value	Within R^2	Obs.
Happy	0.286	0.043	0.000	0.281	312
Very Happy	0.207	0.048	0.000	0.146	312
Very Unhappy	−0.039	0.010	0.000	0.044	312
V. Happy – V. Unhappy	0.246	0.054	0.000	0.160	312
Happiness (mean)	1.063	0.180	0.000	0.245	312
Satisfaction (mean)	2.313	0.420	0.000	0.344	312
Satisfied	0.440	0.076	0.000	0.347	312
Happy + Satisfied	0.366	0.058	0.000	0.389	310

Note: Countries surveyed at least twice. Standard errors clustered by country. On average over the eight indicators, the within-country R^2 is 0.24.

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Table S10: Other country-level correlates of national well-being: Shapley decomposition of R^2 (WVS).

Indicator	Other variable	Obs.	R^2 other alone	Shapley decomposition of R^2			
				Income	Region	Other	Total
Satisfaction (mean)	Median income (PIP, log)	339	0.19	0.11	0.32	0.07	0.50
Satisfaction (mean)	Household consumption p.c. (log)	292	0.30	0.10	0.29	0.13	0.51
Satisfaction (mean)	Freedom of choice	334	0.63	0.11	0.21	0.42	0.74
Satisfaction (mean)	Importance of God	329	0.01	0.17	0.33	0.02	0.51
Satisfaction (mean)	Tolerance (homosexuality)	326	0.24	0.12	0.30	0.10	0.52
Satisfaction (mean)	Importance of democracy	216	0.08	0.13	0.32	0.02	0.47
Satisfaction (mean)	GDP growth (5-year)	334	0.02	0.16	0.34	0.02	0.52
Happiness (mean)	Median income (PIP, log)	339	0.03	0.03	0.32	0.01	0.36
Happiness (mean)	Household consumption p.c. (log)	292	0.08	0.03	0.28	0.04	0.35
Happiness (mean)	Freedom of choice	333	0.39	0.02	0.22	0.27	0.51
Happiness (mean)	Importance of God	330	0.01	0.05	0.31	0.03	0.39
Happiness (mean)	Tolerance (homosexuality)	325	0.09	0.02	0.31	0.07	0.40
Happiness (mean)	Importance of democracy	216	0.03	0.02	0.28	−0.03	0.27
Happiness (mean)	GDP growth (5-year)	334	0.01	0.03	0.31	0.00	0.34

Note: Country-year averages of WVS variables: perceived freedom of choice and control (1–10), importance of God in one's life (1–10), justifiability of homosexuality (1–10), importance of living in a democracy (1–10, waves 5–7), and average annual growth of real GDP per capita over the previous five years. Shapley decomposition of the R^2 of the regression on log GDP per capita, region dummies and the other variable. In each row, the largest contribution among income, region and the other variable is in bold.

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Table S11: Countries most often ranked happiest (WVS, 8 indicators $\times$ 8 samples: each wave and all waves combined).

Country	Occurrences as happiest	Region
Switzerland	10	Western
Mexico	9	Latin America
Kyrgyzstan	6	Asia
Australia	5	Western
Netherlands	4	Western
Nigeria	4	Africa
Puerto Rico	4	Latin America
Colombia	3	Latin America
Finland	3	Western
Vietnam	3	Asia

Note: Looking at all waves combined, the happiest country-year is: Kyrgyzstan–2020 (3 indicators: Very Happy, V. Happy – V. Unhappy, Happiness (mean)); Finland–1981 (Satisfied); Malaysia–2012 (Very Unhappy); Mexico–2012 (Satisfaction (mean)); Qatar–2010 (Happy); Vietnam–2020 (Happy + Satisfied). For *Very Unhappy*, the happiest country-year is the one with the lowest share.

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Table S12: Effects of question wording and scale on life evaluation, Fabre (2025) survey experiment.

Dependent Variables: Model:	Answer		Answer (0–10 equiv.)	Satisfied (6+)	Answer (0–10 equiv.)	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
Ladder wording (Gallup)	-0.451*** (0.046)	-0.455*** (0.067)	-0.456*** (0.067)	-0.088*** (0.013)	-0.464*** (0.123)	-0.494** (0.172)
Scale 1–10	0.137** (0.046)	0.132 (0.068)	-0.260*** (0.072)	-0.008 (0.013)	-0.276*** (0.046)	-0.540** (0.182)
Ladder wording (Gallup) × Scale 1–10		0.008 (0.092)	-0.041 (0.097)	0.028 (0.019)		0.055 (0.247)
Income quartile (1–4)					0.689*** (0.031)	0.635*** (0.043)
Income quartile (1–4) × Ladder wording (Gallup)					-0.004 (0.042)	0.009 (0.058)
Income quartile (1–4) × Scale 1–10						0.106 (0.061)
Income quartile (1–4) × Ladder wording (Gallup) × Scale 1–10						-0.022 (0.083)
<i>Fixed-effects</i>						
Country	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	11,000	11,000	11,000	11,000	11,000	11,000
R ²	0.05681	0.05681	0.05942	0.04208	0.15976	0.16024

Heteroskedasticity-robust standard-errors in parentheses

Signif. Codes: ***: 0.001, **: 0.01, *: 0.05

Note: OLS with country fixed effects, survey weights (normalized within country) and heteroskedasticity-robust standard errors in parentheses. *Ladder wording*: Cantril ladder (vs. WVS life satisfaction). *Scale 1–10*: answers from 1 to 10 (vs. 0 to 10). In columns 3, 5 and 6, 1–10 answers are linearly stretched to 0–10. Columns 5 and 6 interact the treatments with the respondent's income quartile within their country; in column 6, the interaction between income and wording is 0.009 ($p = .883$) and the triple interaction is -0.023 ($p = .786$). * $p < .05$, ** $p < .01$, *** $p < .001$.

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Table S13: Decomposition of the WVS–Gallup gap by country.

Country	Year WVS/EVS (Gallup)	Survey: mode	Gallup (1)	WVS/EVS (2)	Gap D (2)–(1)	Ladder 0–10 (3)	Satisf. 1–10 (4)	Question effect Q (4)–(3)	Residual R $D - Q$
Poland	2017	EVS: CAPI	6.28	7.56	1.28	5.53	6.37	0.84	0.44
Spain	2017	EVS: CAPI	6.47	7.49	1.02	5.61	6.99	1.38	–0.36
Japan	2019	WVS: Mail	5.94	6.76	0.82	5.33	5.00	–0.33	1.15
Italy	2018	EVS: CAPI	6.69	7.28	0.59	5.99	6.59	0.60	–0.01
Germany	2018	WVS: CAPI	7.15	7.74	0.58	6.01	6.84	0.83	–0.25
United Kingdom	2022	WVS: CAPI/Web/Mail	6.76	7.32	0.56	5.67	6.77	1.10	–0.55
France	2018	EVS: CAPI	6.79	7.26	0.47	5.96	6.68	0.72	–0.25
Switzerland	2017	EVS: Web/Mail/CAPI	7.58	8.02	0.43	6.85	7.31	0.46	–0.03
Saudi Arabia	2003 (2006)	WVS: Face-to-face	7.06	7.28	0.23	6.31	7.10	0.79	–0.57
United States	2017	WVS: Web/Phone	7.23	7.22	–0.01	6.03	6.74	0.70	–0.71

Note: Mean answers on native scales (Gallup and ladder: 0–10; WVS/EVS and satisfaction: 1–10); columns (3) and (4): Fabre (2025). The Gallup year is given in parentheses when it differs from the WVS year. Survey: mode: source of the WVS/EVS data and main interview mode(s) (CAPI: computer-assisted personal interviews).

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Table S14: Decomposition of the WVS–Gallup gap: summary.

Statistic	Estimate	95% bootstrap CI
Mean gap D (WVS – Gallup)	0.60	[0.55; 0.64]
Mean question effect Q (satisfaction 1–10 – ladder 0–10)	0.71	[0.56; 0.84]
Mean residual $R = D - Q$ (samples, mode, context)	–0.11	[–0.26; 0.04]
Share of mean gap due to question ($\bar{Q}/\bar{D}$)	1.19	[0.94; 1.45]
Share of gross level difference due to question ($(\bar{Q} /(\bar{Q} + \bar{R}))$)	0.86	[0.76; 0.99]
Share of cross-country variance of D due to Q ($\text{cov}(D, Q)/\text{var}(D)$)	0.11	[–0.33; 0.55]
Correlation between D and Q	0.09	[–0.26; 0.41]
Mean absolute gap $ D $	0.60	[0.56; 0.65]
Mean absolute residual $ R $	0.43	[0.33; 0.60]
Mean gap, share satisfied (6+)	0.07	[0.06; 0.08]
Mean question effect, share satisfied (6+)	0.11	[0.08; 0.14]
Variance share due to Q , share satisfied	0.04	[–0.33; 0.45]

Note: 10 countries. 95% confidence intervals from 1,000 bootstrap replications resampling respondents within country $\times$ variant (Fabre) and within country (WVS). For Gallup, only the distribution of answers is available: in each country, the numbers of respondents giving each answer (0 to 10) are redrawn from a multinomial distribution with the observed shares and sample size, which is equivalent to resampling respondents.

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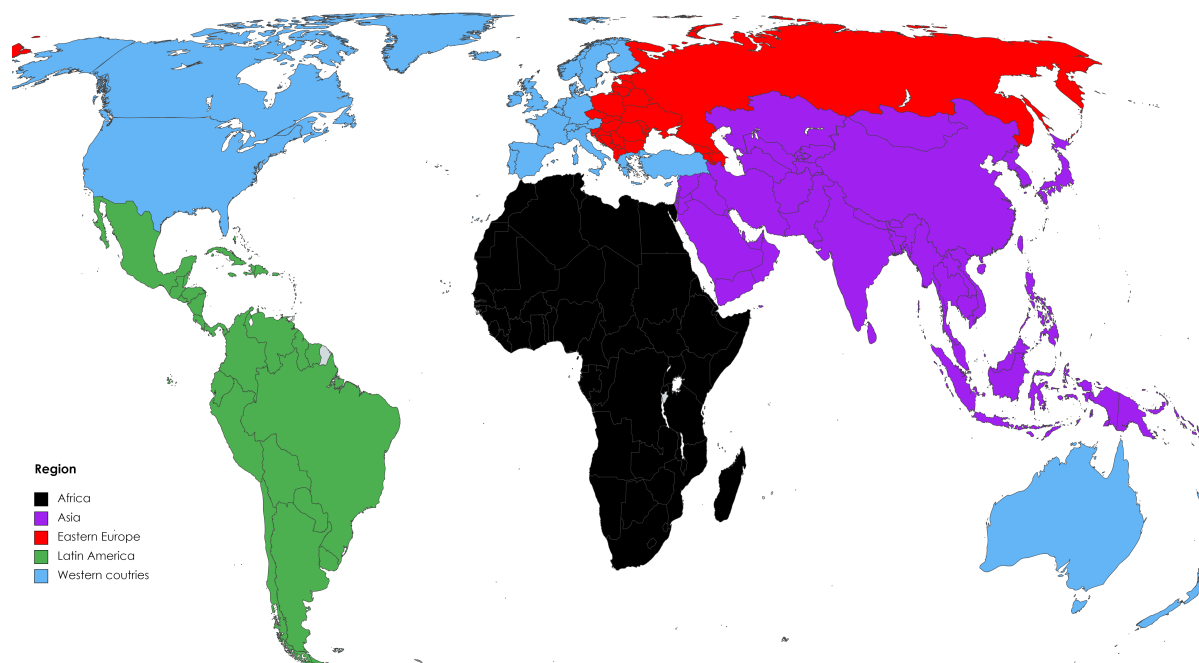
Table S15: Representativeness of WVS/EVS samples: sample shares relative to population benchmarks.

Measure	Surveys	Median				Correlation with log GDP
		All	GDP < 10k	10–30k	> 30k	
Tertiary education, 25+ (ratio sample/benchmark)	295	1.18	1.98	1.18	1.05	–0.40
Urban (ratio sample/benchmark)	76	1.44	2.19	1.45	1.22	–0.69
Aged 65+ (ratio sample/benchmark)	335	0.90	0.64	0.88	1.01	0.33
Women among married respondents – 0.5	337	0.01	–0.00	0.01	0.01	0.19

Note: One observation per WVS or EVS survey. Ratios: weighted sample share divided by the population share, medians across surveys (a ratio above 1 indicates over-representation). Benchmarks (World Bank WDI): share of the population aged 25+ with at least short-cycle tertiary education (UNESCO), or with complete or incomplete tertiary education (25) before 2011; urban population share (national definitions; sample: urban and peri-urban respondents, waves 4, 5 and 7 only); share aged 65+ among the population aged 15+ (sample: 65+ among respondents, usually 18+). Women among married: unweighted share of women among married or cohabiting respondents minus 0.5. Correlations with log GDP per capita (PPP) of the log ratios (or of the deviation).

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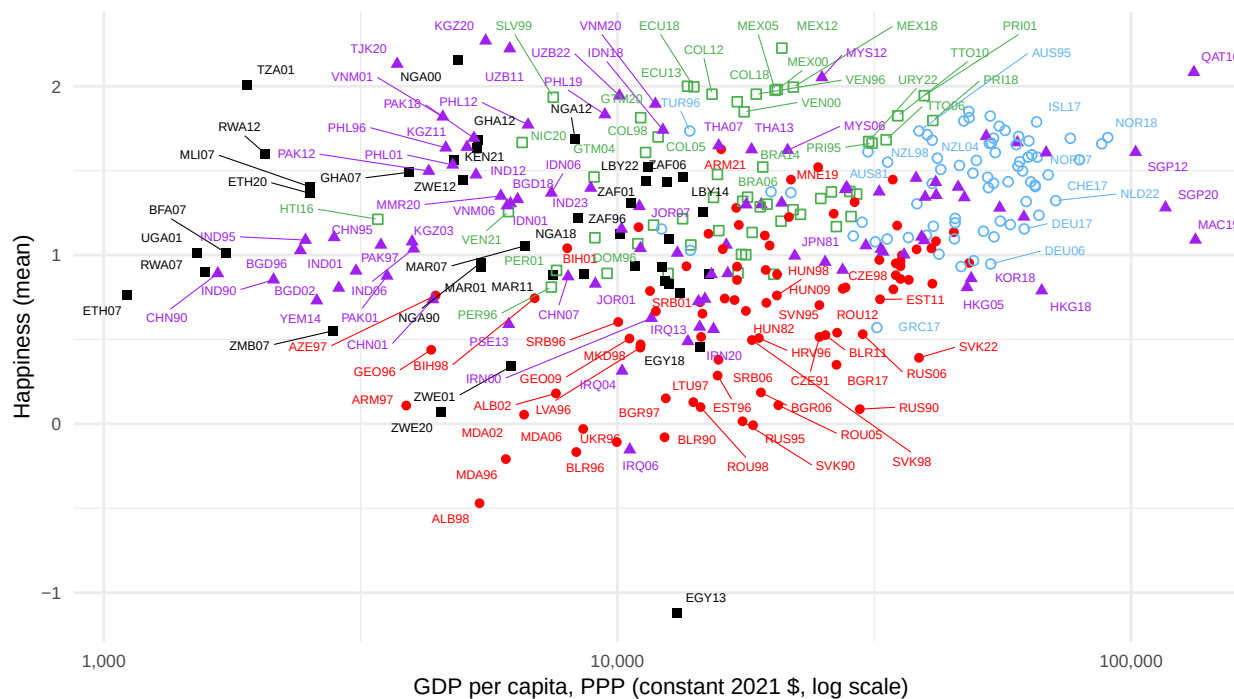
Figure S1: Main classification of countries into five world regions.



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Figure S2: Mean happiness vs. GDP per capita (PPP), WVS/EVS 1981–2023.



R-squared (log GDP) = 0.04

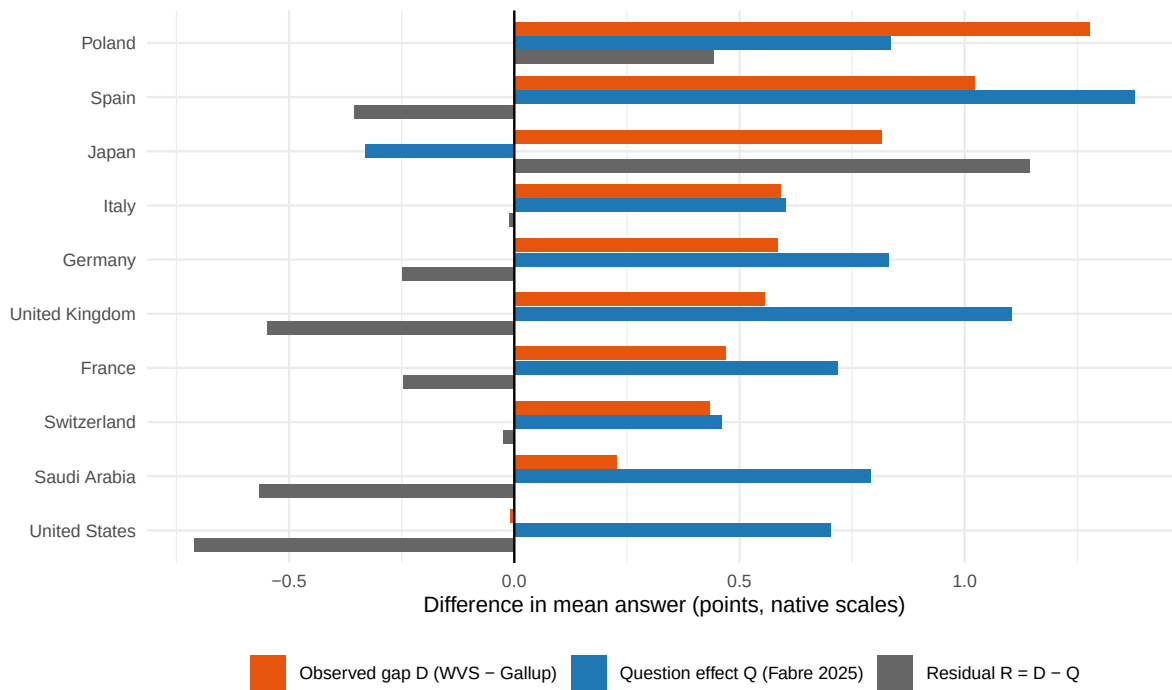
■ Africa ▲ Asia ● Eastern Europe □ Latin America ○ Western

Note: Answers coded +3 (Very happy), +1, -1, -3 (Not at all happy).

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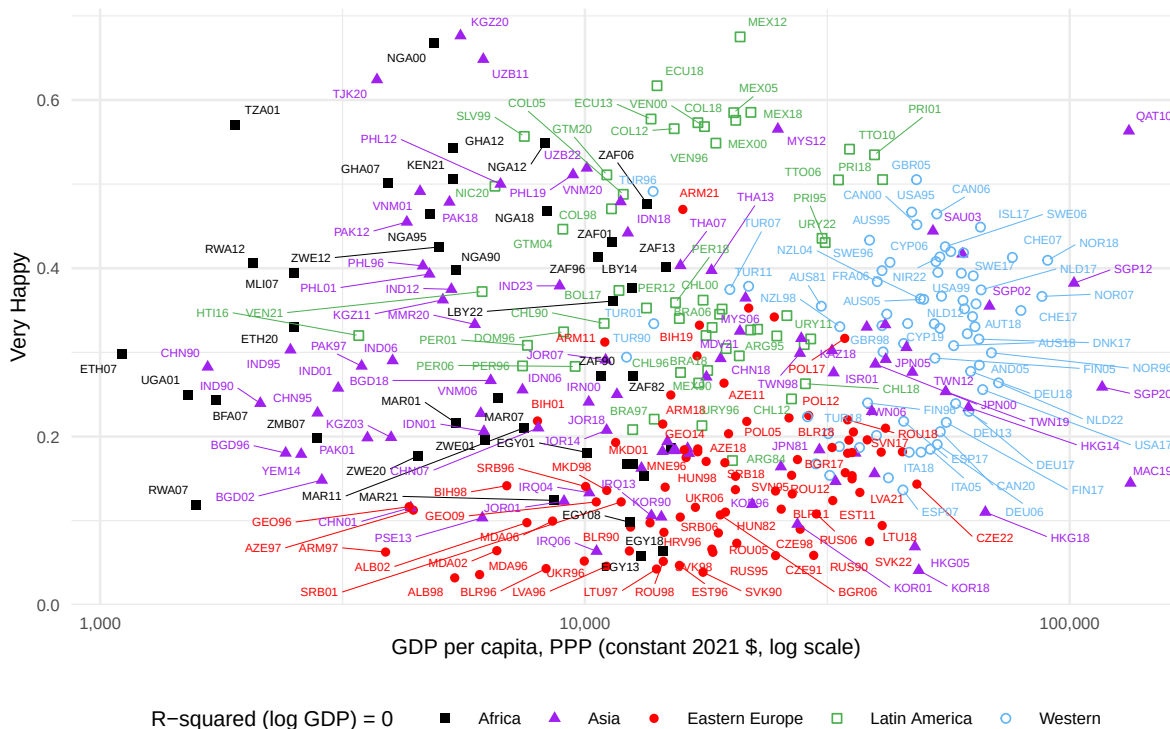
Figure S3: Observed WVS–Gallup gap, question effect and residual by country.



Note: D : mean satisfaction (WVS/EVS, 1–10) minus mean ladder (Gallup, 0–10), same year. Q : mean answer to satisfaction 1–10 minus mean answer to ladder 0–10, Fabre (2025). $R = D - Q$.

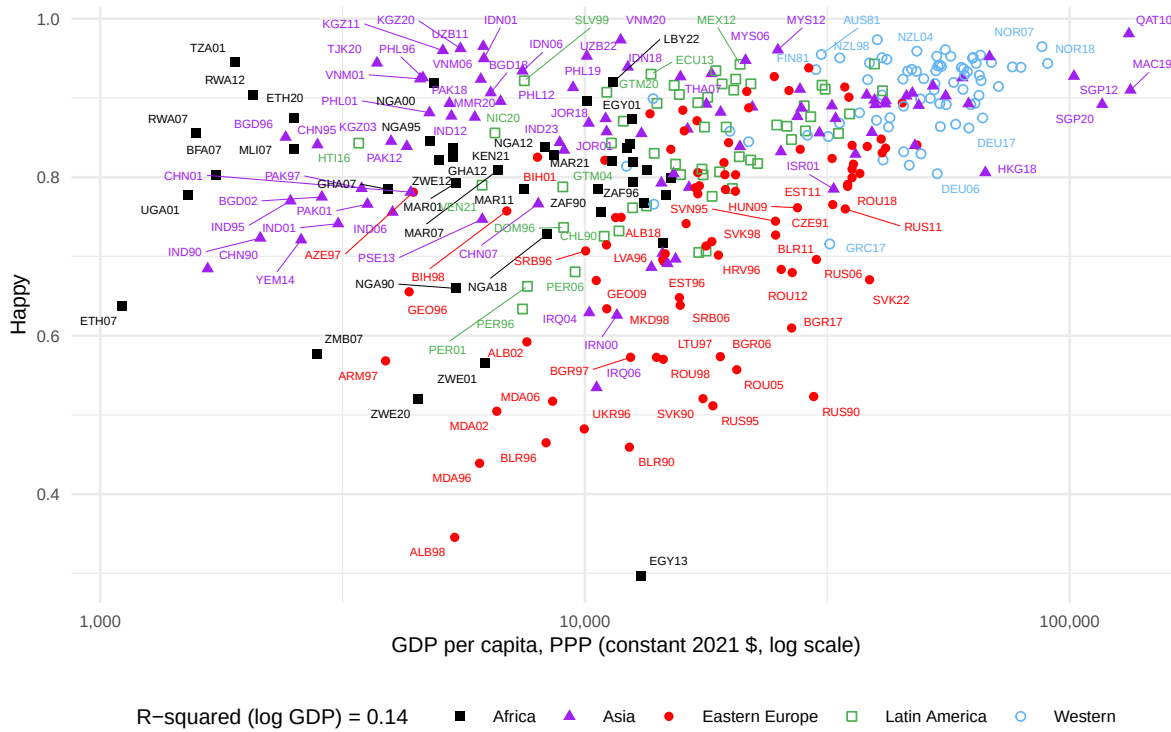
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Figure S4: Share of *Very happy* people vs. GDP per capita (PPP), WVS/EVS 1981–2023.



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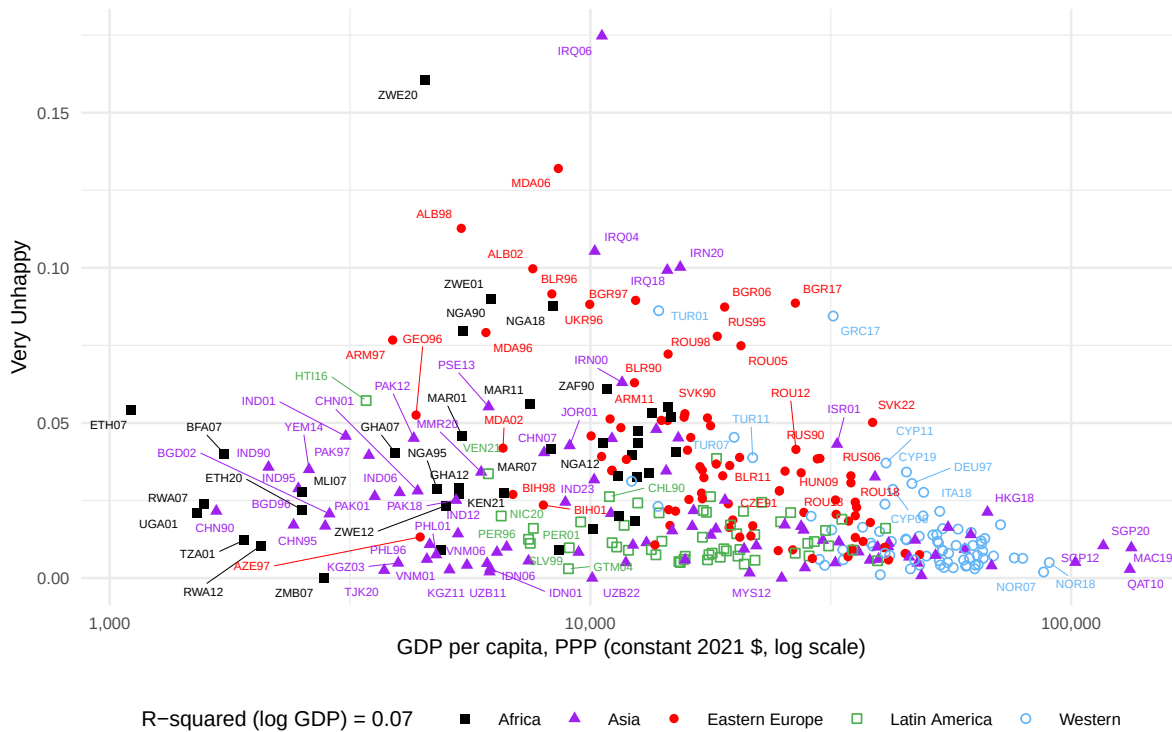
Figure S5: *Happy* vs. GDP per capita (PPP), WVS/EVS 1981–2023.



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Figure S6: *Very Unhappy* vs. GDP per capita (PPP), WVS/EVS 1981–2023.

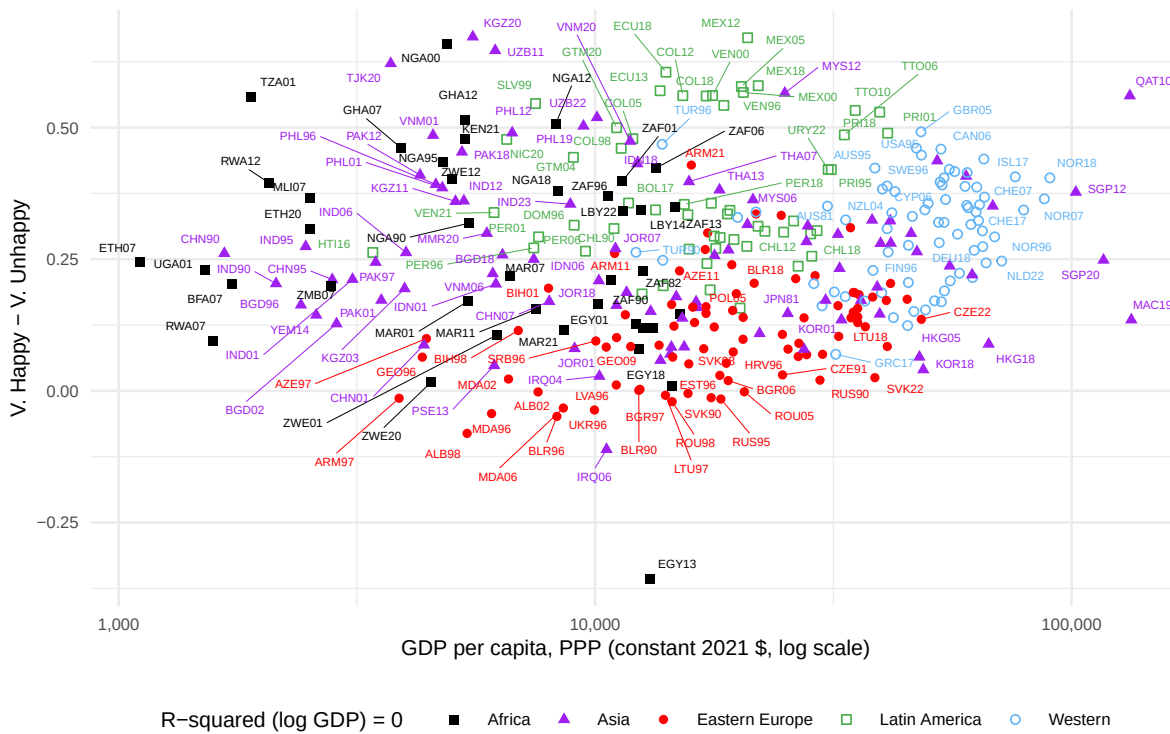


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Note: Egypt 2013 (42% of very unhappy people) is not shown, to keep the y -axis readable; the R^2 in the legend includes it.

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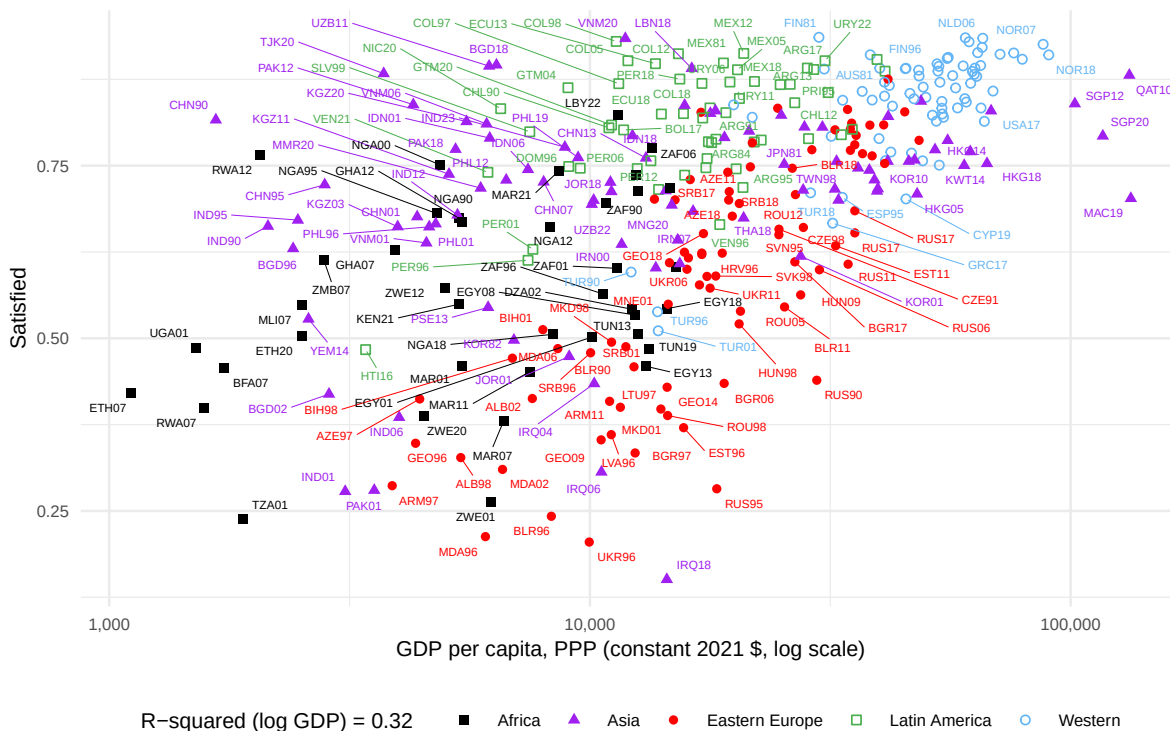
Figure S7: *V. Happy* – *V. Unhappy* vs. GDP per capita (PPP), WVS/EVS 1981–2023.



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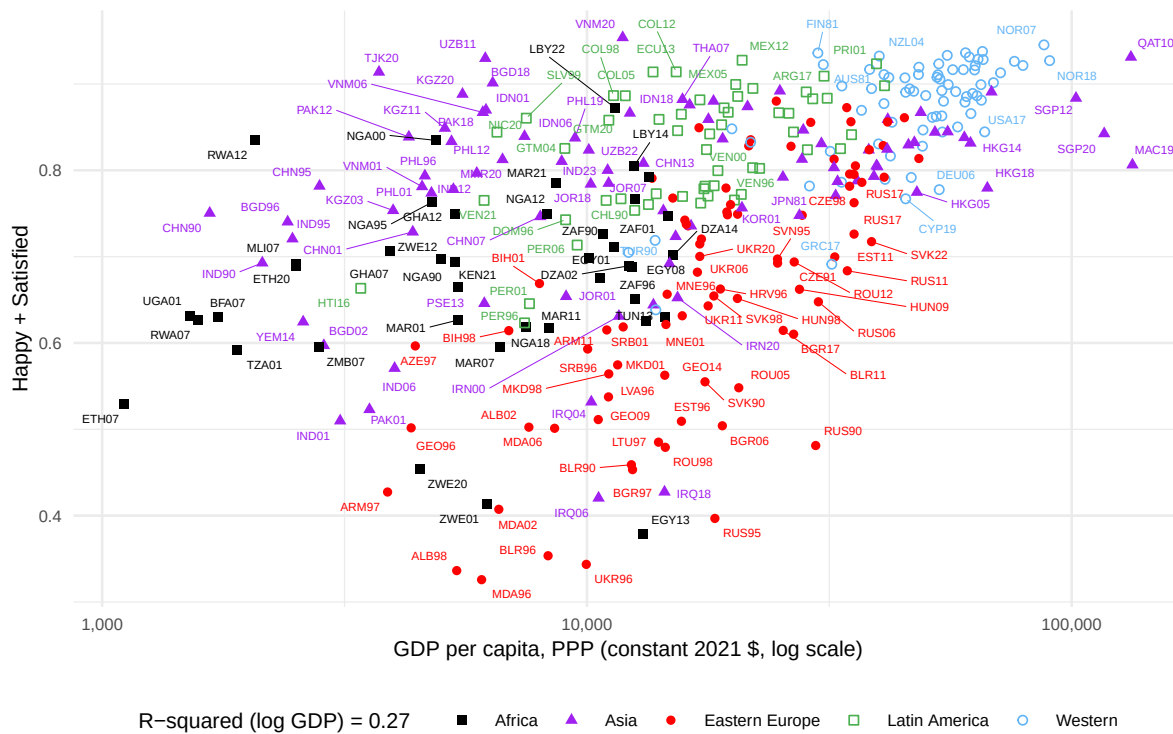
Figure S8: *Satisfied* vs. GDP per capita (PPP), WVS/EVS 1981–2023.



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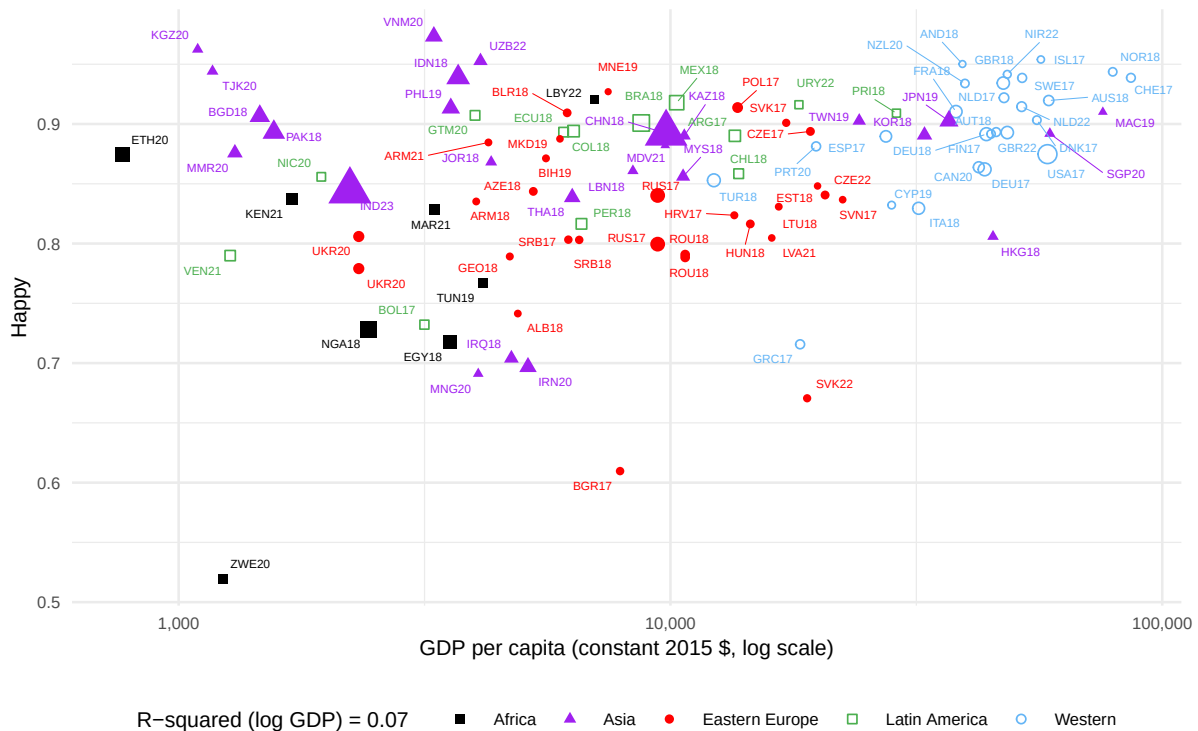
Figure S9: Happy + Satisfied vs. GDP per capita (PPP), WVS/EVS 1981–2023.



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Figure S10: Happy vs. GDP per capita (nominal), WVS/EVS wave 7 (2017–2023), weighted by population.



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Note: Point size and regression weights proportional to population.

References

- [1] Christian Haerpfer, Ronald Inglehart, Alejandro Moreno, Christian Welzel, Kseniya Kizilova, Jaime Diez-Medrano, Marta Lagos, Pippa Norris, Eduard Ponarin, and Bi Puranen. World values survey trend file (1981–2022) cross-national data-set. JD Systems Institute & WVSA Secretariat, Madrid & Vienna. Data file version 3.0.0, 2022. URL <https://doi.org/10.14281/18241.23>.
- [2] EVS/WVS. Joint EVS/WVS 2017–2022 dataset (joint EVS/WVS). GESIS, Cologne, ZA7505, data file version 5.0.0 (2024-06-24), 2024. URL <https://doi.org/10.4232/1.14320>.
- [3] Timothy N. Bond and Kevin Lang. The sad truth about happiness scales. *Journal of Political Economy*, 127(4):1629–1640, 2019. doi: 10.1086/701679. URL <https://doi.org/10.1086/701679>.
- [4] Ronald Inglehart and Hans-Dieter Klingemann. Genes, culture, democracy, and happiness. In Ed Diener and Eunkook M. Suh, editors, *Culture and Subjective Well-Being*, pages 165–183. MIT Press, Cambridge, MA, 2000. doi: 10.7551/mitpress/2242.003.0012. URL <https://doi.org/10.7551/mitpress/2242.003.0012>.
- [5] Gallup. Gallup global research methodology (world poll), 2022. URL https://news.gallup.com/file/poll/165404/World_Poll_Methodology_042722.pdf.
- [6] Hadley Cantril. *The Pattern of Human Concerns*. Rutgers University Press, New Brunswick, NJ, 1965.
- [7] World Happiness Report. World happiness report 2026: Data for figure 2.1 (2011–2025). Accessed 27 September 2026, 2026. URL <https://worldhappiness.report/data-sharing/>.
- [8] World Bank. World development indicators. GDP, population, household consumption, education, urbanization and age-structure indicators, downloaded from the World Bank API on 27 September–1 October 2026, 2026. URL <https://databank.worldbank.org/source/world-development-indicators>.
- [9] International Monetary Fund. World economic outlook database (DataMapper). Indicators PPPPC, NGDPDPC and LP, accessed 27 September 2026, 2026. URL <https://www.imf.org/external/datamapper>.
- [10] Richard H. Lindeman, Peter F. Merenda, and Ruth Z. Gold. *Introduction to Bivariate and Multivariate Analysis*. Scott, Foresman, Glenview, IL, 1980.
- [11] Ulrike Grömping. Estimators of relative importance in linear regression based on variance decomposition. *The American Statistician*, 61(2):139–147, 2007. doi: 10.1198/000313007X188252. URL <https://doi.org/10.1198/000313007X188252>.
- [12] Carol Graham and Eduardo Lora, editors. *Paradox and Perception: Measuring Quality of Life in Latin America*. Brookings Institution Press, Washington, DC, 2009.
- [13] Mariano Rojas, editor. *Handbook of Happiness Research in Latin America*. Springer, Dordrecht, 2016. doi: 10.1007/978-94-017-7203-7. URL <https://doi.org/10.1007/978-94-017-7203-7>.
- [14] Sergei Guriev and Ekaterina Zhuravskaya. (un)happiness in transition. *Journal of Economic Perspectives*, 23(2):143–168, 2009. doi: 10.1257/089533009788430535. URL <https://doi.org/10.1257/089533009788430535>.
- [15] Betsey Stevenson and Justin Wolfers. Economic growth and subjective well-being: Reassessing the Easterlin paradox. *Brookings Papers on Economic Activity*, 2008(Spring):1–87, 2008. doi: 10.1353/eca.0.0001. URL <https://doi.org/10.1353/eca.0.0001>.

- [16] Daniel W. Sacks, Betsey Stevenson, and Justin Wolfers. The new stylized facts about income and subjective well-being. *Emotion*, 12(6):1181–1187, 2012. doi: 10.1037/a0029873. URL <https://doi.org/10.1037/a0029873>.
- [17] Richard A. Easterlin. Does economic growth improve the human lot? some empirical evidence. In Paul A. David and Melvin W. Reder, editors, *Nations and Households in Economic Growth*, pages 89–125. Academic Press, New York, 1974. doi: 10.1016/b978-0-12-205050-3.50008-7. URL <https://doi.org/10.1016/b978-0-12-205050-3.50008-7>.
- [18] Richard A. Easterlin, Laura Angelescu McVey, Malgorzata Switek, Onnicha Sawangfa, and Jacqueline Smith Zweig. The happiness–income paradox revisited. *Proceedings of the National Academy of Sciences*, 107(52):22463–22468, 2010. doi: 10.1073/pnas.1015962107. URL <https://doi.org/10.1073/pnas.1015962107>.
- [19] Angus Deaton. Income, health, and well-being around the world: Evidence from the Gallup World Poll. *Journal of Economic Perspectives*, 22(2):53–72, 2008. doi: 10.1257/jep.22.2.53. URL <https://doi.org/10.1257/jep.22.2.53>.
- [20] Adrien Fabre. Public acceptance of international redistribution in high-income countries. Working paper, 2025. URL https://github.com/bixiou/robustness_global_redistr.
- [21] August Håkan Nilsson, Johannes C. Eichstaedt, Tim Lomas, Andrew Schwartz, and Oscar Kjell. The Cantril ladder elicits thoughts about power and wealth. *Scientific Reports*, 14:2642, 2024. doi: 10.1038/s41598-024-52939-y. URL <https://doi.org/10.1038/s41598-024-52939-y>.
- [22] Gallup. Worldwide research methodology and codebook (updated october 2023). Gallup, Washington, DC, 2023. URL <https://www.unicef.org/innocenti/media/3346/file/Gallup-World-Poll-Methodology-2023.pdf>.
- [23] David G. Blanchflower. Is the Gallup World Poll reliable? After Babel, 2024. URL <https://www.afterbabel.com/p/gallup-world-poll>.
- [24] World Values Survey Association. Rules and procedures for WVSA principal investigators in wave 8, 2023. URL https://www.worldvaluessurvey.org/documents/RULES_AND_PROCEDURES_FOR_WVSA_PIs_IN_WAVE_8.pdf.
- [25] Robert J. Barro and Jong Wha Lee. A new data set of educational attainment in the world, 1950–2010. *Journal of Development Economics*, 104:184–198, 2013. doi: 10.1016/j.jdeveco.2012.10.001. URL <https://doi.org/10.1016/j.jdeveco.2012.10.001>.
- [26] Ulrich Kohler. Surveys from inside: An assessment of unit nonresponse bias with internal criteria. *Survey Research Methods*, 1(2):55–67, 2007. doi: 10.18148/srm/2007.v1i2.75. URL <https://doi.org/10.18148/srm/2007.v1i2.75>.